04 - Insertion Sort and Merge sort Results for Ismailov Jaloliddin (VIHVFL)

ⓘ Correct answers are hidden.

Score for this attempt: **0.7** out of 1
Submitted Oct 6 at 1:52pm
This attempt took 30 minutes.

Question 1

0.4 / 0.4 pts

Complete the pseudo code of Minimum Selection Sort!

Note: Minimum Selection Sort Divides the array into a sorted first part and an unsorted second part. In the beginning, the sorted part is empty. In each pass, it selects the minimum of the unsorted part and exchanges it with the first item of the unsorted part. Thus, in each pass, the length of the sorted part increases by one. It goes on while the unsorted part contains at least two elements.

Example: [3,2,7,8,4,1] => [1|2,7,8,4,3] => [1,2|7,8,4,3] => [1,2,3|8,4,7] => [1,2,3,4|8,7] => [1,2,3,4,7|8] = [1,2,3,4,7,8].

```
minSelSort( A : T [ n ] )
   for( i := 0 to n-2 )
        [ Select ]
        for( [ Select ] )
             if( [ Select ] )
                  ind := j
             else
                  skip
        swap( A[i], A[ind] )
```

Answer 1:

i := 0 to n-2

Answer 2:

ind := n-1

Answer 3:

j := n - 2 downto i

Answer 4:

A[j] < A[ind]

Answer 5:

skip

Question 2

0.1 / 0.1 pts

Which of the following statements are true?

- ✓ The time complexity of insertion sort in the best case is $\Theta(n)$.
- ☐ The running time of insertion sort is always the same when the input sizes are equal.
- ✓ The time complexity of insertion sort in the worst case is $\Theta(n^2)$.
- ✓ Merge sort breaks down a problem into multiple sub-problems and solves them recursively.
- ☐ The time complexity of merge sort is $\Theta(n \log n)$ in the worst case and $\Theta(n)$ in the best case.
- ☐ In insertion sort the end of the array becomes sorted first.
- ✓ Insertion sort works in-place.

Partial

Question 3

0.03 / 0.1 pts

Perform the insertion sort on the following array: A = [15, 19, 6, 8, 11]

Give the number of key comparisons (between the elements of the array):

9

Consider the data movements of the elements of the array.

Give the number of these data movements: 14

Give the number of loop iterations: 10

Answer 1:

9

Answer 2:

14

Answer 3:

10

Partial

Question 4

0.17 / 0.2 pts

Perform the merge sort on the following array: A = [36, 27, 12, 24, 32, 15, 22, 35, 10].

Give the results of each sorted merge in order. Please enter the values separated by comma, without any whitespace or bracket.

1. 27,36

2. 12, 24

3. 12,24,27,36

4. 15,32

5. 10,35

6. 10,22,35

7. 10,15,22,32,35

8. 10,12,15,22,24,27,3

Answer 1:

12, 24

Answer 2:

12,24,27,36

Answer 3:

15,32

Answer 4:

10,35

Answer 5:

10,22,35

Answer 6:

10,15,22,32,35

Answer 7:

10,12,15,22,24,27,32,35,36

Incorrect

Question 5

0 / 0.2 pts

Perform the insertion sort on the following, 0-indexed array:

A = [24, 9, 2, 10, 19, 28, 24, 12]

Give the initial sorted subarray and the current state of it after every insertion. Please enter the values separated by comma, without any whitespace or bracket.

Initial sorted subarray: A[0..0] = _____

The sorted subarray after the 2nd iteration: A[0..2] = _____

The sorted subarray after the 5th iteration: A[0..5] = _____

Give the number of key comparisons and data movements at each iteration:

During the 2nd iteration: key comparisons: _____ , data movements: _____

During the 5th iteration: key comparisons: _____ , data movements : _____

Answer 1:

(You left this blank)

Answer 2:

(You left this blank)

Answer 3:

(You left this blank)

Answer 4:

(You left this blank)

Answer 5:

(You left this blank)

Answer 6:

(You left this blank)

Answer 7:

(You left this blank)

Quiz Score: **0.7** out of 1